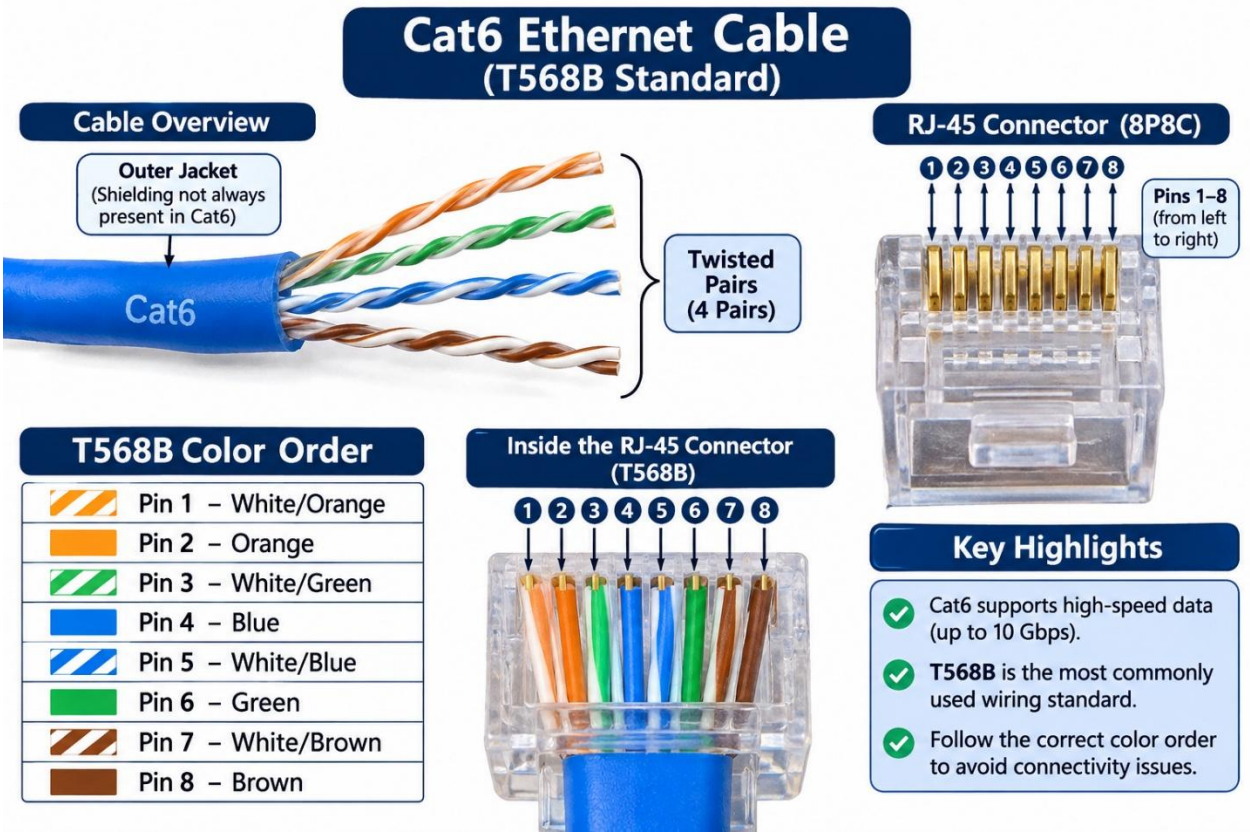
Network Cable and connectors

Tasks 1 : Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.



Key Components:

- **Cat6 cable:** Contains four twisted pairs designed for Ethernet data transmission.
- **RJ-45-style connector (8P8C):** Terminates the cable and connects it to an Ethernet port.
- **Eight pins/contacts:** Make electrical contact with the eight conductors.
- **Twisted pairs:** Help reduce electromagnetic interference and crosstalk.

Conclusion: Correct T568B wiring helps ensure reliable Ethernet connectivity. Both ends wired to T568B make a straight-through cable.

Task 2 : List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

1. Cat5e Ethernet cable

Connector: RJ-45-style 8P8C

Used to connect a home Wi-Fi router to a desktop PC or a Zomato POS terminal. It supports Gigabit Ethernet over a standard 100 m channel.



2. Cat6 Ethernet cable

Connector: RJ-45-style 8P8C

Suitable for gaming PCs, network switches, and IPL screening consoles. It offers better crosstalk performance than Cat5e and can support 10 Gbps over shorter distances under suitable conditions.



3. Fiber-optic cable

Connectors: LC or SC, depending on equipment

Used to connect data-center switches or network equipment across buildings. It carries data as light and is suitable for high bandwidth and long-distance links.



Cable Type	Example Application or Device	Why It Is Used
Cat5e Ethernet	Home Wi-Fi router connected to a POS terminal	Supports reliable Gigabit Ethernet for ordinary network traffic.
Cat6 Ethernet	Gaming PC or IPL screening console connected to a switch	Provides good performance and reduced crosstalk for high-speed local connections.
Fiber-optic cable	Data-center switches or links between buildings	Supports high bandwidth over long distances and is resistant to electromagnetic interference.

Conclusion: The correct cable depends on the required speed, distance, environment, and compatibility of the connected devices.

Task 3 : Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Tools and Materials

- Cat5e or Cat6 Ethernet cable
- Compatible RJ-45-style 8P8C connectors
- Cable stripper or crimping tool
- Wire cutter
- Ethernet cable tester

Step-by-Step Procedure

1. **Measure and cut:** Measure the required cable length and cut it cleanly.
2. **Strip the outer jacket:** Carefully remove a short section of the outer jacket without damaging the internal conductors.
3. **Separate the wire pairs:** Untwist and arrange the eight conductors according to the T568B standard.
4. **Arrange the colours:** White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
5. **Trim evenly:** Hold the wires in order and cut their ends straight so they are all the same length.
6. **Insert into the connector:** Push the conductors into the correct channels of the RJ-45-style connector. Check that all eight reach the front and that the outer jacket enters far enough to be held by the connector.
7. **Crimp the connector:** Place the connector into the compatible crimping tool and press firmly to secure the contacts and cable.
8. **Terminate the other end:** Repeat the process using T568B at the other end to create a straight-through cable.
9. **Test the cable:** Use an Ethernet cable tester to check that all eight conductors are connected in the correct order.

Two Common Mistakes to Avoid

1. **Incorrect colour sequence:** Mixing up the wire order can cause the cable to fail or operate incorrectly.
2. **Wires not fully inserted:** If one or more conductors do not reach the contacts, the connection may fail. Always inspect the plug before crimping.

Conclusion

Correct wire ordering, careful termination, and cable testing are essential for a reliable Ethernet connection.

Task 4 : Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Task 4: Selecting Network Cables for an IPL Screening Gaming Zone

Scenario

A gaming zone is being set up for an IPL screening event. Multiple gaming consoles, screens, and network devices must be connected reliably.

Recommended Cable and Connector Types

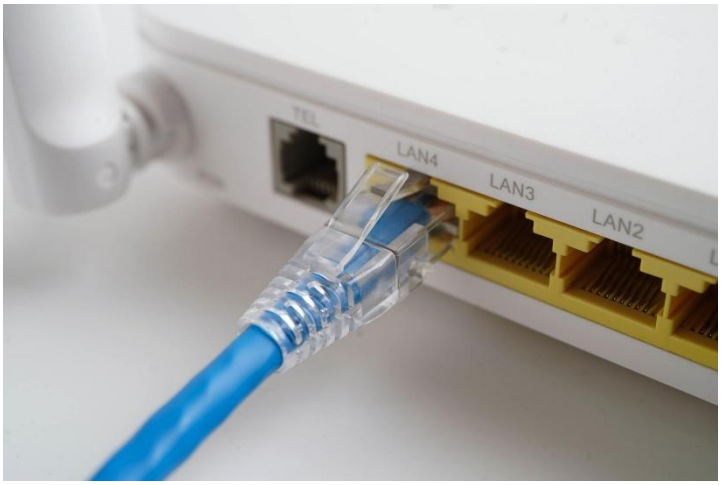
Connection	Recommended Cable	Connector	Reason
Gaming consoles or PCs to network switch	Cat6 Ethernet	RJ-45-style 8P8C	Provides reliable, low-latency wired networking for online gaming.
Smart TVs or streaming devices to network switch	Cat6 Ethernet	RJ-45-style 8P8C	Provides a stable connection for streaming IPL matches and reduces dependence on Wi-Fi.
Main network switch to router	Cat6 Ethernet or a compatible higher-speed Ethernet link	Compatible Ethernet connector	Connects the local gaming network to the internet gateway.
Long-distance connection between buildings or network racks	Fiber-optic cable	Compatible LC or SC connector	Supports high bandwidth over long distances and avoids electromagnetic interference.

Explanation

Cat6 Ethernet is a practical choice for connecting nearby gaming consoles, PCs, and smart TVs because wired connections generally provide more consistent latency and reliability than Wi-Fi. A network switch can connect multiple wired devices, while a router provides access to the internet. Fiber-optic cable is suitable for longer backbone links when supported by compatible network equipment.

Task 5 : Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Photo:



Cable Type: Ethernet twisted-pair cable (Cat5e or Cat6, depending on the cable markings).

Connector Type: RJ-45-style 8P8C connector.

Devices Connected: The cable connects a router's LAN port to a wired device, such as a desktop computer, network switch, or POS terminal.

Purpose: It carries network data between the connected devices and provides a stable wired connection.

Conclusion: Ethernet cables are widely used in home and business networks because they provide reliable wired communication between network devices.